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Key uncertainties behind global projections of direct air capture deployment

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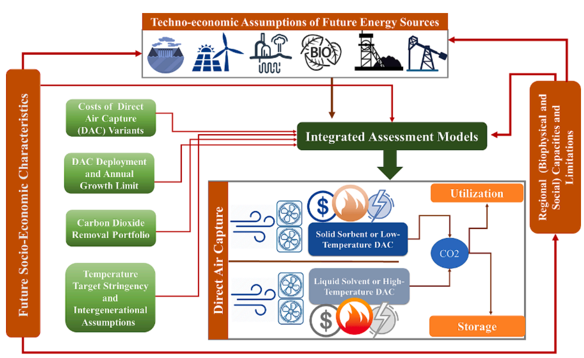
HIGHLIGHTS

- Rich portfolios of carbon dioxide removal methods lead to more sustainable modeling outcomes.
- Socio-economic pathways significantly affect the direct air capture deployment.
- National studies can provide better decision support in a direct air capture context.
- Integrated assessment models should include different variants of direct air capture.
- Both storage and utilization options should be available in modeling studies.

GRAPHICAL ABSTRACT



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ABSTRACT

To limit global warming to levels set in the Paris Agreement, integrated assessment modeling highlights the necessity of removing gigatons of carbon dioxide. Direct air capture is one of the potential methods for carbon dioxide removal and has gained significant attention despite being a relatively new technology. While some scenarios can meet Paris targets without relying on direct air capture, other models show the need for up to 38 gigatons/yr of removal by this technology. Such wide variation in projections reflects uncertainty over the feasibility of deploying direct air capture at scale and points the need for greater attention to the model assumptions driving this variation. This study provides a comprehensive review of scenarios from the Intergovernmental Panel on Climate Change Sixth Assessment Report and peer-reviewed studies focusing on direct air capture as a mitigation strategy. The review identifies several key factors contributing to uncertainty in direct air capture projections, including prolonged fossil fuel use, diversity of carbon removal methods, future socio-

Abbreviations: IAM, Integrated Assessment Model; GHG, Greenhouse Gas; CO_2, Carbon Dioxide; CDR, Carbon Dioxide Removal; IPCC, International Panel on Climate Change; AR6, Sixth Assessment Report; AR, Afforestation/Reforestation; BECCS, Bioenergy with Carbon Capture and Storage; DAC, Direct Air Capture; HT-DAC, High-Temperature Direct Air Capture; LT-DAC, Low-Temperature Direct Air Capture; EOR, Enhanced Oil Recovery; DACCU, Direct Air Carbon Capture and Utilization; DACCS, Direct Air Carbon Capture and Storage; GDP, Gross Domestic Product; SSP, Shared Socio-Economic Pathway; SDR, Social Discount Rate.

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economic conditions, intergenerational assumptions, deployment limitations, and technology costs. The study recommends conducting multi-model assessments of direct air capture scenarios due to the influence of various assumptions and model structures on projections. The authors also suggest including different variants of direct air capture technology and both carbon utilization and storage pathways in future models, as these factors can impact system demands, economic efficiency, and public acceptability.

1. Introduction: The necessity of carbon dioxide removal and roles for direct air capture

The Paris Agreement identified the objective of limiting global warming to well below 2 °C (end-of-century temperature compared to the pre-industrial level) and pursuing limits to 1.5 °C to prevent the dangerous impacts of climate change [1]. To find the pathways compatible with these targets, energy-economy modelers take advantage of integrated assessment models (IAMs), which are designed to capture the interdependencies, system linkages and feedback between biophysical systems and socio-economic characteristics of possible futures to identify trajectories of greenhouse gas emissions (GHG) [2]. Different IAMs have investigated pathways that may plausibly limit increases in global warming to levels of less than 2 or 1.5 °C at the end of the century [3,4]. Within these pathways, carbon dioxide removal (CDR) plays an increasingly important role.

CDR refers to processes (practices or technologies) that reduce the concentration of CO_2 in the atmosphere [6–8]. The effect of CDR on atmospheric levels of CO_2 is similar to emission reductions and may be preferred where removals of CO_2 can be achieved at a lower cost than emission reductions. CDR may be beneficial for offsetting residual emissions in hard-to-abate sectors or non-CO_2 emissions [3]. In addition, many pathways consistent with the Paris Agreement targets project that global emissions should be balanced between emissions and removals of greenhouse gases between 2050 and 2070 [5,9,10]. However, these pathways may have a period where the Paris Agreement temperature targets are exceeded – a condition referred to as temperature overshoot. Atmospheric CO_2 concentrations are then drawn down through subsequent deployment of CDR at net-negative levels (net removals exceed net emissions).

CDR takes a wide variety of forms, and while there are different CDR categorization schemes [11], one approach clusters different methods into nature and technology-based ones. The former consists of practices that try to enhance the natural processes that uptake CO_2 (e.g., soil carbon management and enhanced weathering) and methods such as afforestation/reforestation (AR) that remove CO_2 by changing land use [12]. Two well-established technology-based methods are bioenergy with carbon capture and storage (BECCS) and direct air capture (DAC). CDR methods differ in their environmental footprints and sustainability implications, technological readiness, the feasibility of deployment, potential, storage permanency, costs and public acceptability [13,14]. Despite the significant reliance on CDR in pathways meeting the Paris Agreement targets, the current deployment scales for most novel methods (e.g., DAC and BECCS) are negligible [15].

This study focuses on DAC as a relatively new technology with uncertain high costs and energy requirements. DAC captures CO_2 by binding it to sequestration materials, followed by a regeneration step, where the captured carbon is released, and the sequestration materials are prepared for reuse. Capturing materials can be in liquid (solvent) or solid (sorbent) form. DAC systems based on solvents typically require a higher temperature heat energy than solid sorbents [16]. Table 1 shows that these two DAC variants' system demands, costs, and environmental footprints vary substantially. Further details about the capturing process and system demands of DAC are provided in Section S1 of the supplementary information. Even with the current high costs and energy requirements, DAC appears to have some characteristics, such as large-scale CO_2 removal potential, siting flexibility (although limited by geological storage and availability of low-carbon energy sources), low direct land and water footprint (compared to BECCS and AR), carbon storage permanency and high measurement certainty, that has attracted the attention of policy-makers, industries and academics [17–19].

The captured carbon from DAC may be utilized for various applications or stored in geological reservoirs. Direct air carbon capture and utilization (DACCU) might have different carbon balance implications and climate change effects from direct air carbon capture and storage (DACCS). DACCS is a "negative emissions" system from a carbon balance standpoint. However, depending on the utilization pathway, DACCU might only be carbon neutral, where, for example, the captured carbon is being used to produce carbon-based fuels [23]. In this paper, our primary focus will be on DACCS, as it has been predominantly examined in modeling studies, with only a few exceptions where DACCU has also been considered [24].

The deployment rate of DACCS in pathways meeting the Paris Agreement temperature targets varies by multiple orders of magnitude among different modeling studies, covering the range of 0 to 18.9 $GtCO_2/year_{2100}$ for such scenarios available in the AR6 database [25]. The top end of these projections rises to around 39 $GtCO2/year_{2100}$ when looking at scenarios from other peer-reviewed studies that investigate the role of DACCS. The substantial variation in DACCS deployment rates across different modeling studies presents a challenge to the fundamental role of IAMs in informing decision-makers. Despite notable studies on DACCS, the existing literature needs a comprehensive review that consolidates the scattered information from various studies and investigates the plausible reasons behind such variations. This paper addresses this research gap by conducting an in-depth analysis of modeling assumptions and findings. Furthermore, this study proposes best modeling practices for future research by incorporating insights from different studies and disciplines. By focusing our attention on the central drivers influencing DACCS deployment, this paper enables better identification of realistic assumptions and a better understanding of pathways' social, environmental, and technical implications, including DACCS and other CDR methods, which will be essential to generate

Table 1
Characteristics of two main variants of DAC technology.

System Demands of Direct Air Capture	Direct Air Capture Technology Variants	
	HT-DAC	LT-DAC*
Required Temperature for Thermal Energy (°C)	800–900 [16]	80–120[16]
Thermal Energy Requirements (GJ/tCO_2)	7.7–10.7[16]	3.4–4.8[16]
Required Electricity (GJ/tCO_2)	0.74–1.7[16]	0.55–1.1 [16]
Water Demand (tH2O/tCO2)[a]	5[20]	3[20]
Land for 1Gt CO2 removal (10^3 Km^2/year)[b]	20[20]	11.5[20]
Required Sorbent/Solvent (Mt) for 1 Gt CO_2 removal	6[20]	3[20]
Material Use (Mt) for 1 Gt CO_2 removal	17[20]	36[20]
Capture Costs (\$/tCO_2)[c]	147–279[16]	88–228[16]

a. Water demand for BECCS is 300 to 750 (tH_2O/tCO_2) [21].
b. The land requirement for 1 Gt CO_2 removal by BECCS is around 300–1100 (10^3 Km^2/ year) [21].
c. These capture costs do not consider the CO_2 emitted for supplying the energy requirements. Detailed cost analysis is available at Fasihi et al. (2019) and the National Academies of Sciences, Engineering, and Medicine (2019).
* LT-DAC has different variants, and the system demands shown in this table are for a system called "Temperature Swing Adsorption or TSA"[22].

better models, communicate the embedded uncertainties regarding DACCS deployment and enhance the interpretability of the results by readers and decision-makers [18,25–33].

To draw conclusions about what factors are most influential on DACCS deployment, this article examines two complementary scenario sources: (1) aggregated information from the AR6 database [25] and (2) individual published IAM studies (all but one of them being available in the AR6 database) that include more detailed information on socio-technical and economic modeling assumptions for DACCS than is available in the first set of scenarios [18,26–33]. The AR6 scenarios database allows for a comparison of a large number (700) of scenarios with different compositions of CDR. The AR6 scenario database only hosts the outputs of scenario studies, and the underlying assumptions (including study design and model structure) of modeling studies matter. We, therefore, consulted peer-reviewed studies of DACCS deployment to capture dimensions that cannot be determined from the AR6 database alone. This set of nine studies is summarized in Table 3 and provides in aggregate, 54 scenarios that offer greater transparency and detail on key CDR-related assumptions for further review in this article. By comparing the scenarios which differ in critical dimensions, we can make observations concerning the influence of these factors, which provides insight into significant uncertainties.

The paper proceeds as follows. Section 2 analyzes the AR6 database regarding total CDR deployment and narrows the focus to DACCS and its interaction with BECCS and AR. Section 3 introduces and examines the 54 DACCS scenarios, discussing the key factors that drive significant variation among model projections. We describe the differences found in the identified studies in four categories: 1) uncertainties regarding socio-economic futures, 2) intergenerational assumptions, 3) maximum capacity and growth rate limitations and 4) costs of DACCS. Drawing on Sections 2 and 3, Section 4 analyses the areas that appear likely to be valuable for further investigation to better model and realize the DACCS system. The conclusion section summarizes the main findings from this study, its limitations and suggests directions for future research.

2. Lessons from the AR6 scenario database

The AR6 scenarios database categorizes scenarios based on the climate change target they are designed to meet. We review four categories of these pathways: C1, C2, C3 and C4. These deep mitigation scenarios maintain the temperature increase below 1.5 or 2 °C [28,34], with variations in probability and overshoot. On average, these four categories not only use CDR as negative emissions but also, compared to scenarios meeting less stringent mitigation targets than 2 °C, include a higher share of low-carbon energy, a higher electricity share in final energy, less CO_2 intensity of primary energy, less energy demand and fewer non-energy GHG emissions [4]. Depending on the temperature goal and probability of overshoot, the timing of net-zero global CO2 emissions will occur around 2055 to 2085. After reaching net zero, net-negative emissions are between 1 and 10 $GtCO_2$ (the range is subject to temperature goal and overshoot) to compensate for the earlier positive emissions [4].

In addition to the effects of temperature targets on CDR deployment, models use various technical and socio-economic assumptions as inputs or constraints to produce their results as well as socio-economic factors, such as population and gross domestic product (GDP). Fig. 1 shows a significant variation in GDP|PPP and population assumptions among scenarios. These factors strongly influence future global emissions [35], subsequently affecting the scale at which CDR must be deployed. In the new generations of scenarios, scenario producers have introduced the shared socio-economic pathways (SSPs) concept that depicts five worlds with different socio-economic characteristics [36]. The population [37] and GDP [38] projections associated with the five SSP worlds are also shown in Fig. 1 to compare the values that modeling studies (ones that meet below 2 °C target) are using against what is available in the SSP database [39]. The effects of SSPs on DACCS deployment are analyzed below in section 3.1 in more detail.

In addition to the future's socio-economic characteristics, energy-related techno-economic factors increase or decrease the need for CDR. Models might use these factors as constraints (e.g., the potential for biomass energy) or assumptions (e.g., the future cost of renewable energy technologies), ultimately affecting CDR deployment. Since the models' inputs (assumptions and constraints) are not available in the AR6 database, we use the outputs as proxies to extract high-level insights from scenarios in the database. We limited this analysis to 212 scenarios meeting the 1.5 °C end-of-century target (C1 and C2 categories) to exclude the changes that differences in the remaining carbon budget might imply. We also include data for non-biomass renewable and fossil fuel energy supply and final energy demand for comparison. Since the estimates of CO_2 sequestered specifically by AR is not available for many scenarios (though it is embedded in the Agriculture, Forestry, and Other Land Use or AFOLU emissions), BECCS, DACCS and enhanced weathering are used to calculate the CDR deployment. Based on the distribution of cumulative CDR deployment (2050–2100), two groups of scenarios with high (equal to or above the 75th quantile) and low (equal to or lower than the 25th quantile) CDR deployment are determined. Fig. 2 shows how these two groups vary regarding the final energy demand, energy supply by fossil fuels and renewables, and probability of temperature overshoot.

Among the aforementioned scenario factors that might drive differences in projections for cumulative CDR deployment, only projections of fossil fuel energy supply show a clear difference. Scenarios with low CDR

Table 2
Categories of deep mitigation scenarios from the AR6 database [4,25].

Scenario Category	Number of Simulated Scenarios	Median Cumulative CDR Deployment ($GtCO_2$)	Number of scenarios that include DACCS	Median [min–max] DACCS Deployment ($GtCO_2$/year)		
				2050	2080	2100
C1	97	659	38	0.3 [0–2.2]	0.8 [0.1–8.8]	2.2 [0.1–13.8]
C2	133	687	33	0.2 [0–2.2]	3.8 [0–20.3]	7 [0–18.9]
C3	311	569	125	0.3 [0, 2]	2.9 [0.1–9.8]	3.4 [0.1–11]
C4	159	553	57	0.3 [0–1.6]	3.1 [0–9.7]	2.4 [0–21]

C1: 1.5 °C end-of-century temperature goal (a probability of greater than 0.5) and no or limited overshoot (probability of peak warming greater than 1.5 °C less than 0.67).
C2: 1.5 °C end-of-century temperature goal (a probability of greater than 0.5), but allowing for high overshoot (probability of peak warming greater than 1.5 °C ≥ 0.67).
C3: Likely below 2 °C temperature limit for the whole century (a probability of greater than 0.67).
C4: Below 2 °C end-of-century temperature goal (a probability of greater than 0.5).

Table 3
Reviewed integrated assessment modeling studies.

Study	Goal and Insights
Marcucci et al. (2017)	• Investigation of the role of DACCS in meeting the PA targets and their effects on the energy transition and other system demands. Recognition of DACCS as an essential technology in making the achievement of 1.5 °C possible and reducing mitigation costs. The deployment projections are 38.2 $GtCO_2/year_{2100}$ for 1.5 °C and 21.3 $GtCO_2/year_{2100}$ for 2 °C scenarios.
Holz et al. (2018)	• Evaluating the role of CDR in achieving the 1.5 °C target and its trade-offs with stringent emissions reduction. Emphasizing the significance of enhancing near-term emissions reductions of pre-2030 NDC targets. Highlighting the issue of intergenerational equity by examining the implications of incorporating CDR in reducing near-term ambitions. Projecting the deployment of DACCS at approximately 3.5 $GtCO_2/year_{2100}$.
Strefler et al. (2018)	• Assessment of different scenarios with varying temperature targets (1.5 and 2 °C) and limitations on total CDR deployment (e.g., 8, 12 and 20 $GtCO_2$/year). The rising transitional challenges of the economy and energy sector when total CDR is less than 5 $GtCO_2$/year. The need for at least 8 $GtCO_2$/year CDR to meet 1.5 ∘C target and more than 15$GtCO_2$/year to keep transitional challenges manageable. A significant variation of DACCS deployment from 0 to 10.8 GtCO2 depending on the temperature target and limitations on total CDR.
Realmonte et al. (2019)	• The uncertainties associated with biological CDR methods and the need to consider other methods, such as DACCS, to meet the PA targets. Investigating the role of DACCS under different cost estimates, energy requirements, maximum capacity ($GtCO_2$/year) assumptions, and growth rates (%). Temperature overshoot of up to 0.8 °C in case of relying on DACCS without subsequent deployment. A projection of DACCS deployment around 30 and 28 $GtCO_2/year_{2100}$ to achieve a goal of 1.5 and 2 °C targets, respectively.
Fuhrman et al. (2020)	• Investigation of the role of DACCS in meeting the 1.5 target and its implications for food, water, and energy systems. Introducing DACCS into models to some extent reduces the rate of increase in food prices compared to those scenarios relying only on BECCS and AR. Highlighting the need for an urgent reduction in emissions to avoid the adverse land–water-energy effects introduced by the large-scale deployment of CDR methods. DACCS deployment from 10 to 29 $GtCO_2/year_{2100,}$ depending on its future cost trajectory and permitted overshoot level.
Fuhrman et al. (2021)	• Investigation of DACCS role under different shared socio-economic pathways (SSPs) in meeting the 2 and 1.5 °C goals. The possibility of a high deployment of DACCS, especially when climate policies are delayed or significant challenges in mitigating emissions exist. Depending on the socio-economic trajectory and the temperature target, DACCS might play a role from 0.03 to 31 $GtCO_2/year_{2100}$.
Grant et al. (2021),	• Modeling and analyzing the phenomenon of mitigation deterrence where the high future deployment of CDR in models dilutes near-term mitigation actions. Highlighting the benefits of separating CDR and emissions reduction targets in climate strategies. A projection of CO_2 removal by DACCS in the range of 10.1 to 20.53 $GtCO2/year_{2100}$.
Strefler et al. (2021a)	• Investigating the effects of different carbon pricing mechanisms on CDR deployment under 1.5 and 2 °C goals. Highlighting that alternative carbon prices can reduce long-term economic efforts, limit the total CDR deployment and be more implementable. The varying range of DACCS deployment in 2100 from 21.3 $MtCO_2$ to 7.55 $GtCO_{2,}$ depending on the carbon price trajectory and temperature target.
Strefler et al. (2021b)	• Investigating a comprehensive portfolio of CDR methods to compare transitional challenges (the pace at which emissions decrease) and economic efficiency (GDP loss). The association of lowest transitional challenges with scenarios, including all CDR methods. Range of DACCS deployment between 1.4 and 23 $MtCO_2/year_{2100}$ depending on the temperature target.

deployment show a higher pace of transition from fossil fuels, which is in line with previous studies that show having CDR in the model on a large scale lessens the burdens of transition costs and decelerates the shift from fossil fuels [30,32,40]. The uncertainty range for the interactions of non-biomass renewables and CDR deployment is high because these two sectors might have two possible synergies depending on the CDR method that contributes most to the total CDR deployment. For example, in the case of BECCS, as it also provides energy in addition to carbon removal, we might anticipate that a high deployment of it decreases the need for non-biomass renewables. In contrast, DACCS, as an energy-intensive technology, might need a higher share of renewables in the energy system to reduce the carbon emitted in the energy supply process [41,42].

While most models assume a rise in energy demand and try to supply it, some studies focus on the demand side of the equation by producing narratives supporting future low-energy demand pathways that emit less GHG and therefore do not rely on CDR significantly [43,44]. It should be noted that since scenarios are from different modeling studies, there might be various techno-economic assumptions behind models that also contribute to the significant ranges of uncertainty shown in Fig. 2, which will be discussed further below in section 4.4.

Narrowing down our analysis to DACCS, we find that out of the 700 deep mitigation scenarios, only 254 scenarios had DACCS as an option. One of the reasons behind the underrepresentation of DACCS in scenarios meeting targets below 2 °C (C1, C2, C3 and C4 categories) in the AR6 database is the fact that DACCS is a relatively new technology (compared to BECCS) and not an option in 6 out of 20 global models used to produce the AR6 [25,45]. In addition, there are 189 scenarios in which models had DACCS as an option but were not considered (i.e., activated as an option) in the modeling exercise. In contrast, BECCS has been demonstrated in more than 94% of scenarios. Not having DACCS as a CDR option in many deep mitigation scenarios leads to higher estimates of other CDR methods, such as BECCS [46,47]. Figure S1 of the supplementary information section shows how the statistics of BECCS deployment in 2100 vary in scenarios with DACCS as an option compared to those without DACCS. When scenarios do not include DACCS, the median and average values of BECCS deployment will increase from 8 and 9.6 Gt $CO2/year_{2100}$ to 9.6 and 10.2 Gt $CO2/year_{2100}$.

A high amount of BECCS deployment, however, might be infeasible or impose severe negative impacts on food prices and biodiversity [46,47]. This concern justifies the need to model possible CDR methods with realistic assumptions and reasonably precautionary limitations to avoid unrealistic representations. Yet, we found that many AR6 scenarios over-rely on BECCS and AR to meet climate change targets. We illustrate this problem by comparing projections in the AR6 database with the sustainable potentials of each CDR method provided by Fuss et al. (2018). Fuss et al. (2018) estimated the sustainable deployment potential of different net-negative emission technologies (NETs) concerning their costs; impacts on socio-economic, environmental, and biophysical sectors; and the permanency of sequestration. Plausible sustainable deployment potentials were found for BECCS (0.5 to 5 GtCO2/year) and AR (0.5 to 3.6 GtCO2/year), limited mainly by land availability considerations and biodiversity losses. Using the upper deployment estimations of these two technologies (5 and 3.6 GtCO2/year for BECCS and AR, respectively), we found that most of the 1.5 °C scenarios (C1 and C2 categories from the AR6 database) exceed this

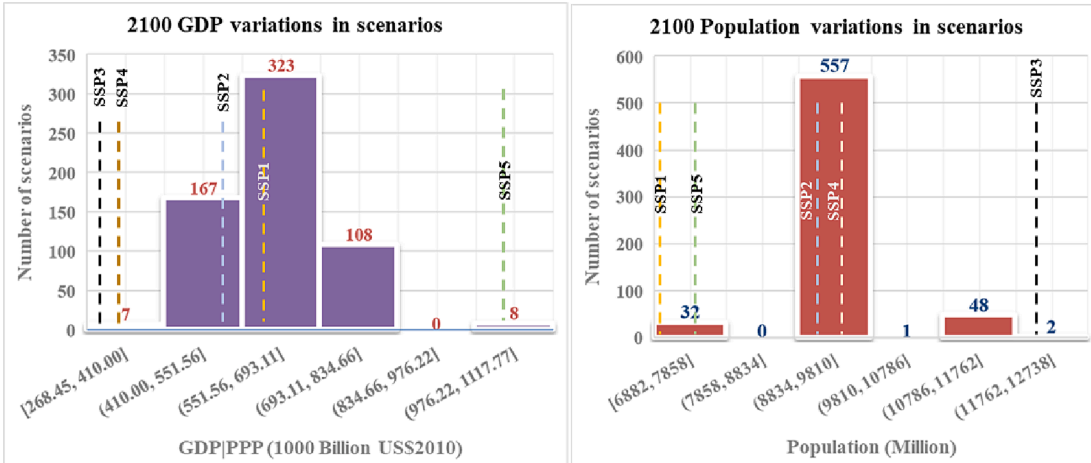

Fig. 1. Distribution of GDP|PPP and population among AR6 deep mitigation scenarios (data from the IIASA AR6 [25] and the SSP database [37–39]).

threshold. Fig. 3 represents both the sustainable potential thresholds [13] and the AR6 [25] modeling results. More than 60% and 85% of scenarios have exceeded the sustainable potential deployment thresholds of AR and BECCS.

Fuss et al. (2018) also provided the potential for DACCS deployment in 2050 (0.5 to 5 GtCO2/year), which was limited by its costs, current storage challenges and potentially unexpected environmental consequences. Importantly, they also determined that if current storage challenges and/or environmental concerns were to be addressed, DACCS might be able to remove up to 40 GtCO2/year in 2100 [13]. Regarding the sustainability implications of CDR methods, Honneger et al. (2021) assessed the impacts of different CDR methods on the relevant sustainable development goals (SDGs). They found the high costs and energy requirements associated with DACCS as the factors dominating the effects of this technology on SDGs which could also limit its deployment potential. They discussed how siting decisions (e.g., deploying DACCS in areas with available waste heat) could alleviate the possible energy-related negative impacts of DACCS on SDGs [14].

As we discussed, the availability of DACCS seems to reduce the reliance on BECCS (also shown in Fig S1). These types of inter-method synergies are not limited to DACCS and BECCS, and that is why studies suggest modeling a rich portfolio of CDR technologies [48–50]. DACCS can also be influenced in the same way by the introduction of new carbon removal methods to models. The extent of this effect depends on various factors, such as the cost of the method, its potential, and the permanency of carbon sequestration [13]. Since five models in AR6 did not have AR as an available CDR method, we could analyze the changes in DACCS deployment between scenarios with AR and those without it. Statistically, we found that the median value of cumulative DACCS deployment in scenarios with AR as an option is lower (around 70%) than those without AR (Figure S2 of the supplementary information section). Accurate and more reliable insights on the effects of technology portfolio diversity on DACCS deployment would benefit from a multi-scenario analysis within the same modeling structure (e.g., one model run including AR and one model run excluding AR). We discuss the advantages of such studies in Section 4.4.

3. Key uncertainties affecting projections of direct air capture deployment under different IAM studies

Since the AR6 database does not provide full information on the underlying assumptions informing the scenarios, we identified nine peer-reviewed studies that generated DACCS scenarios that met the Paris targets. There is a high degree of overlap between the scenarios, including in the AR6 database and the studies chosen, with one study, Realmonte et al. (2019), not part of the AR6 scenarios database. The objectives and critical findings from each study are summarized in Table 3. Within these studies, we identified 54 scenarios for analysis. The key attributes of each scenario analyzed are described in Table 4. Across the scenarios, deployment pathways for DACCS reflect significant differences in scale (between 0 and 38 $GTCO_2/year_{2100)}$ and deployment time. Comparing scenarios that differ in relation to a key variable allowed us to track the influence of these factors on DACCS deployment patterns.

We classify the elements affecting DACCS deployment into change categories shown in Table 5. The variations of DACCS deployment from one scenario to another are only due to the changes stated in the "category of change" column, which was found by comparing the scenarios in each study among themselves (not with scenarios from other studies). Overall, we can attribute the observed uncertainties to four clusters 1) socio-economic futures: different socio-economic narratives that can shape future society and their corresponding greenhouse gas emission activity [36], 2) intergenerational assumptions: stringency of mitigation target, permitted temperature overshoot level and social discount rate 3) maximum capacity and growth rate limitations: whether there is a limit on CDR/DAC deployment and 4) costs of DAC: Both current and future costs of the technology. The subsections below discuss these factors and their importance in DACCS/CDR modeling exercises.

3.1. Uncertainties regarding socio-economic futures

Depending on how the future unfolds over the coming decades, the world may have characteristics that facilitate or hinder the deployment of varying CDR methods. Here "the future" broadly refers to portfolios of socio-economic uncertainties, including population growth rates, economic growth, energy demand, energy supply, and political priorities. In climate change research, the global SSPs package five alternative futures that can be used to explore their potential effects on CDR deployment and DACCS [51] The SSPs contrast different levels of socio-economic challenges to mitigation and adaptation. The variation between challenges results from differences in combinations of underlying assumptions for megatrends of scenario drivers such as population and gross domestic product (shown in Fig. 1), education, technological development, and attitudes toward prioritizing sustainability [36,52].

SSPs affect DACCS deployment in two ways. First, total emissions vary among SSPs for baseline scenarios [39], which has implications for the different amounts of carbon removal that might be required. Second, some SSPs have attributes that lead to increased or decreased DACCS deployments. For example, the SSP1 world depicts a low-carbon and low-energy-demand economy with considerable progress in renewable energy. While the availability of low-carbon energy can be seen as an

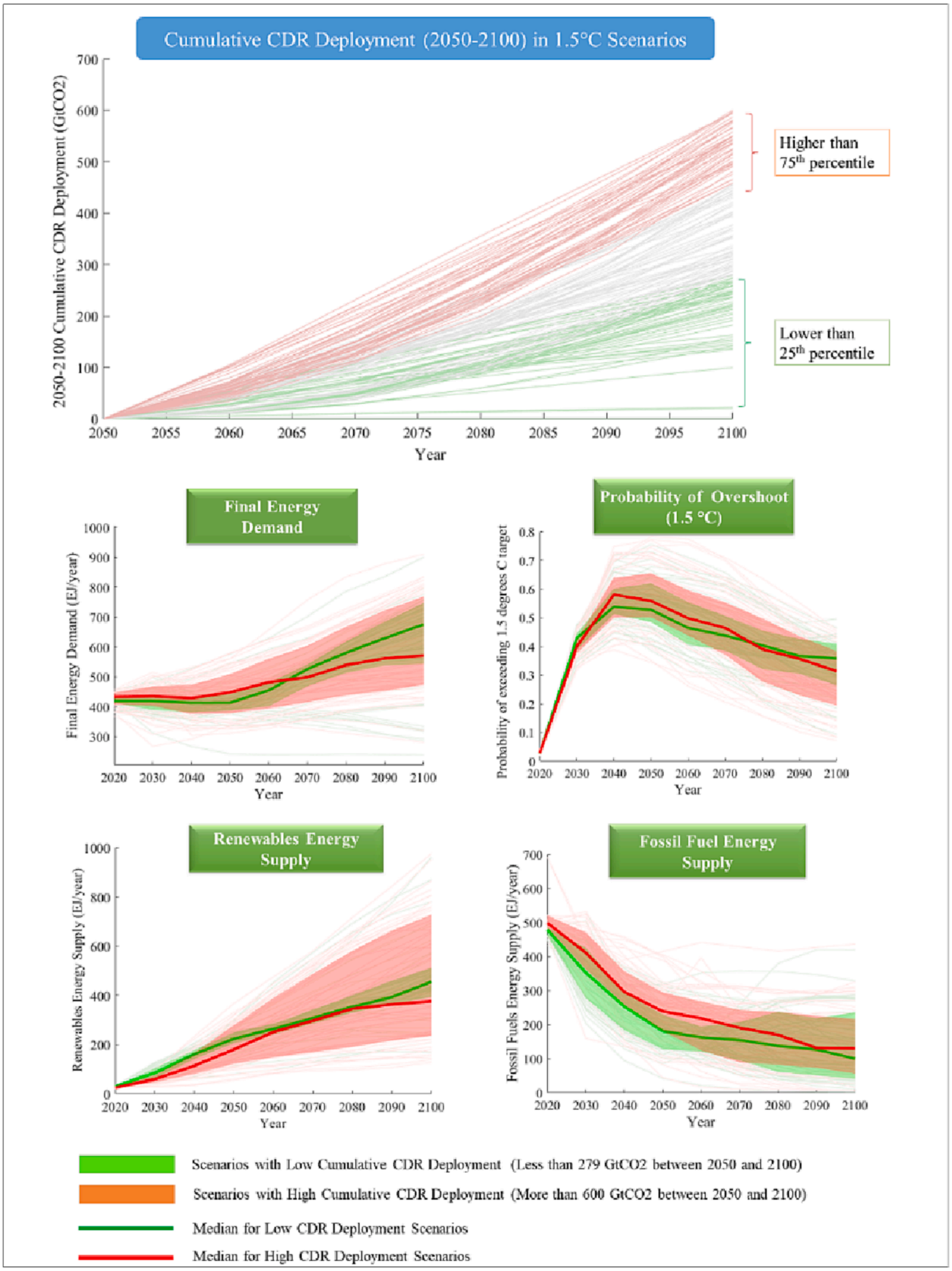

Fig. 2. A comparison of 1.5 °C scenarios (C1 and C2) with high (above 600 $GtCO_2$) and low (lower than 279 $GtCO_2$) CDR deployment in the second half of the century.

opportunity for DACCS to be deployed, people's attitudes toward carbon removal technology might be unfavorable. In SSP1, the priority is to meet sustainability goals through nature-based solutions rather than technological approaches such as DACCS [51]. Some of the characteristics of SSPs that might affect the scale of DACCS deployment are provided in Table S1 and S2 (in the supplementary information of the paper), which are collected from the article introducing the SSP narratives [36] and their scenario elements and five marker scenarios that quantify them with IAMs [53–57]. The SSPs and marker scenarios were developed to give modelers contrasting yet comparable parameterizations (or, in the cases of variables modeled endogenously, target outcomes) for alternative long-term worlds. When SSPs are utilized to project demand for DAC (or other CDR methods), modelers should consider the effects of alternative SSPs (e.g., differences in policy environments) on the technology's costs, energy, and water requirements in addition to its maximum growth (e.g., due to tech-related characteristics and levels of social acceptance). Moreover, differences in factors such as international trade, globalization and technology transfer might have implications for the regional distribution of the technology that models project (see Table S1 in the supplementary information). As shown in Table 4 and Fig. 1, in most of the IAM studies reviewed here, the role of different SSPs is missing, as most studies utilize assumptions for SSP2.

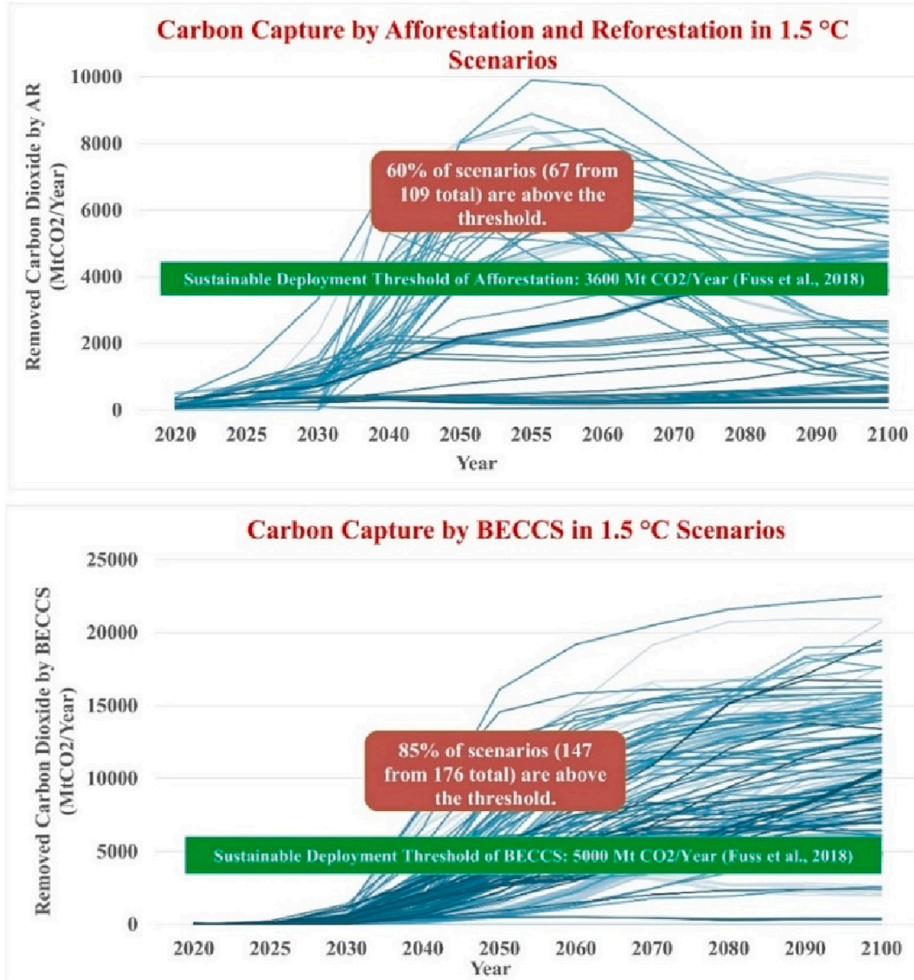

Fig. 3. Afforestation and reforestation (AR) and BECCS (bioenergy with carbon capture and storage) deployment in 1.5 °C scenarios (C1 and C2 categories) of the AR6 database [25] compared to the sustainable potentials suggested by Fuss et al. (2018).

A notable contribution in this area is Fuhrman et al. (2021). They investigated the roles of different CDR methods under various SSPs in scenarios meeting 1.5 and 2 °C targets. In their analysis, many socio-economic and technological factors varied between SSPs, including timing and stringency differences of land-use and climate policy (i.e., carbon price), social acceptance of DACCS technologies (low for SSP1 and intermediate for SSP5; applied by varying the shares of global GDP assigned to DACCS), and energy and cost requirements of DACCS (both the initial value and future trends). Their model could not find any SSP2 pathway that could meet the 1.5 °C goal; also, for SSP3, meeting both 1.5 and 2 °C was impossible. For the 1.5 °C target, Fuhrman et al. (2021) found that an SSP1 (sustainable development) world relies much less on negative emission technologies, and DACCS has a minimal role (1.8 GtCO2/y in 2100). On the other hand, in a world like SSP5, with high energy demand and high fossil fuel combustion, the DACCS role will increase to 30 GtCO2/y in 2050 [29]. The variations between the DACCS deployment rate among different SSPs highlight the importance of incorporating socio-economic scenario uncertainty in the modeling process. Having a lower DACCS deployment scenario under an SSP1 future is also shown in the results of Strefler et al. (2021b), where DACCS deployment decreases from 32.6 (S35) to 1.4 Mt $CO_2/year_{2100}$ (S36) when shifting from the SSP2 to SSP1 scenario [33]. It should be noted that Strefler et al. (2018) also did a sensitivity analysis on SSPs by comparing the SSP2 scenarios with SSP1 and SSP5 futures. These particular scenarios are not available in the IIASA database, but their graphical results are provided in their peer-reviewed study [27].

3.2. Intergenerational Assumptions: Time value of costs and appetite for overshoot

Another key factor determining how much DACCS (and other CDR methods) will be deployed is the timing and level of ambition that scenarios are trying to meet. Such characteristics reflect intergenerational assumptions and affect whether and how much overshoot (and, in turn, DACCS deployment) is acceptable. The social (intergenerational) discount rate (SDR) is applied to future benefits and costs to calculate net present values [58]. The value of the SDR is a critical parameter in determining the deployment scale of DACCS and other CDR methods. Higher values of SDR in the context of climate change policy-making discount the costs of climate action and the impacts of climate change on future generations, thus implying a lower amount of investment in emissions reductions and CDR today. It should also be noted that multiple studies have looked at the SDR from an ethical perspective and discussed the appropriate selection of this value in cost-benefit analysis when intergenerational assets are involved, and topics such as intergenerational equity have been raised [58–61]. In 2015, more than 200 economists participated in a survey recommending what SDR is ethical and reasonable to use in climate change studies. Over 90% of them found SDR values between 1 and 3% acceptable [62]. In the studies that we analyzed in this article, there are very different assumptions regarding the SDR. Marcucci et al. (2017), Strefler et al. (2018) and Strefler et al. (2021) use 5%. Realmonte et al. (2019) changed the SDR from 5% to 0.1%, making the cumulative DACCS deployment fall from

Table 4
Scenarios including DACCS selected from peer-reviewed studies for further analysis and their DACCS-related characteristics.

Study	Integrated Assessment Model	Scenario Number	2100 Temperature Target and Oversoot Level	Socioeconomic		DAC growth rate limit (%/year)	Constraint (Gt CO2/year)	Direct air Capture characteristics				DACCS Deployment (MtCO2/year)			Cumulative DACCS Deployment (GtCO2)
				SSP	Discount Rate			DAC Technology Variants	Cost of DAC ($/tCO2)	Heat GJ/tCO2	Electricity GJ/tCO2	2050	2080	2100	
Marcucci et al. (2017)	MERGE-ETL 6.0	S1	1.5°C	SSP2	5.0%	NF	NF	HT-DAC	600-200	8.1 -5.5	1.8	0.00	10889.70	38259.10	579.17
		S2	2°C		5.0%	NF						0.00	5413.50	21363.50	305.16
Holz et al. (2018)	C-ROADS-5.005	S3	1.5°C	SSP2	NF	NF	2.5 (for each of BECCS, Biochar, Mineralization); 3.5 (DAC)	NF				1668.56	3349.52	3480.11	163.95
	En-ROADS-96	S4		SSP2		NF						1662.35	3343.31	3478.68	163.53
Strefler et al. (2018)	REMIND 1.7	S5		SSP2	5.0%	NF	12 (CDR)	HT-DAC	102.5	10	2	0.19	833.38	3549.23	47.60
		S6				NF	20 (CDR)					1414.55	8393.57	10779.25	361.30
		S7				NF	8 (CDR)					1.34	75.13	2110.03	21.80
		S8	2°C			NF	12 (CDR)					0.00	107.46	1281.90	12.39
		S9	2°C			NF	20 (CDR)					97.52	7371.16	9284.92	255.20
		S10	2°C			NF	8 (CDR)					0.00	0.00	0.00	0.00
Realmonte et al. (2019)	TIAM-Grantham	S11	1.5°C	SSP2	5.0%	20.0%	30 (DAC)-10 (BECCS)	LT-DAC (DAC1) and two HT-DACCS: DAC2 (supplied by waste heat) and DAC21 (heat made on purpose)	DAC1: 300 -100 DAC2/21: 350-50	DAC1: 8.1 DAC2/21: 7.2	DAC1: 1.8 DAC2/21: 1.2	NF			~870
	WITCH	S12	1.5°C	SSP2	3.5%	20.0%									~822
	TIAM-Grantham	S13	2°C	SSP2	5.0%	20.0%									~756
	WITCH	S14	2°C	SSP2	3.5%	20.0%									~812
	TIAM-Grantham	S15	1.5°C	SSP2	5.0%	15.0%	3 (DAC) -10 (BECCS)								~100
		S16	1.5°C	SSP2	5.0%	20.0%									~125
		S17	1.5°C	SSP2	5.0%	30.0%									~140
		S18	1.5°C	SSP2	5.0%	15.0%	30 (DAC)-10 (BECCS)								~380
		S19	1.5°C	SSP2	5.0%	30.0%									~1030
		S20	1.5°C	SSP2	5.0%	20.0%			DAC1: 180-100 DAC2/21: 200-50						~880
		S21	1.5°C	SSP2	0.1%	20.0%			DAC1: 300 -100 and DAC2/21: 350-50						~750
		S22	1.5°C	SSP2	5.0%	20.0%				DAC1: 5.3 DAC2/21: 4.4	DAC1: 1.3 DAC2/21: 0.6				~870
		S23	1.5°C	SSP2	5.0%	20.0%	30 (DAC)-10 (BECCS) +Limited Storage			DAC1: 8.1 DAC2/21: 7.2	DAC1: 1.8 DAC2/21: 1.2				~820
Fuhrman et al. (2020)	GCAM 5.2	S24	1.5°C (High Overshoot)	SSP2	3.00%	NF	NF	HT-DAC	300	8.1	1.8	5685.30	21101.32	23291.35	863.00
		S25	1.5°C (High Overshoot)	SSP2		NF	NF		180	5.3	1.3	12327.89	26779.84	29096.16	1274.95
		S26	1.5°C (Low Overshoot)	SSP2		NF	NF		300	8.1	1.8	5761.55	10015.87	10521.76	497.22
		S27	1.5°C (Low Overshoot)	SSP2		NF	NF		180	5.3	1.3	12029.55	15503.02	16137.06	883.39

~ For scenarios of S11 to S23, yearly DACCS deployment data was not available in the IIASA database, and the authors estimated them based on the figures and data available in the main text of the corresponding study.
NF: Information regarding this element was not found in the study.

Study	Integrated Assessment Model	Scenario Number	2100 Temperature Target	Socio-economic			Direct air Capture characteristics				DACCS Deployment (MtCO2/year)			Cumulative DACCS Deployment (GtCO2)
				SSP	Discount Rate	Carbon Price Path	DAC Technology Variants	Cost of DAC ($/tCO2)	Heat GJ/tCO2	Electricity GJ/tCO2	2050	2080	2100	
Strefler et al. (2021a)	REMIND 2.1	S28	1.5°C	SSP2	5.0%	GDP Growth Rate	HT-DAC	102.5	10	2	1970.33	3580.99	6110.32	162.16
		S29	1.5°C	SSP2	5.0%	Discount rate untill net zero and then constant		102.5	10	2	2.33	5940.10	7552.29	285.81
		S30	2°C	SSP2	5.0%			102.5	10	2	1042.92	583.43	2452.82	33.83
		S31	1.5°C	SSP2	5.0%	Linear (Endegenously) to meet the target		102.5	10	2	0.00	5178.80	7294.05	236.66
		S32	2°C	SSP2	5.0%			102.5	10	2	51.23	126.25	1410.31	13.67
		S33	1.5°C	SSP2	5.0%	Optimal		102.5	10	2	0.00	7.05	21.39	1.61
		S34	2°C	SSP2	5.0%	Optimal		102.5	10	2	0.00	0.96	8.39	0.11
Strefler et al. (2021b)	REMIND-MAgPIE 2.1-4.2	S35	1.5°C	SSP2	5.0%	Constant after 2045	HT-DAC	102.5	10	2	855.42	297.44	32.62	13.89
		S36	1.5°C	SSP1	5.0%			102.5	10	2	23.77	1.38	1.47	0.57
		S37	2°C	SSP2	5.0%	Optimal		102.5	10	2	1.88	1.56	1.47	0.10
Fuhrman et al. (2021)	GCAM 5.3	S38	2°C	SSP4	3.00%	Endegenously to meet the end-of-century target	1.HT-DAC (with natural gas) [DAC11]; 2. HT-DAC (fully electric) [DAC12];3. LT-DAC (electric heat pump) [DAC2]	DAC11: 296 DAC12: 384 DAC2: 402	DAC11: 5.3	DAC11: 1.3 DAC12: 5 DAC2: 2.5	0.45	30.00	30.00	3.35
		S39	2°C	SSP4	3.00%			DAC11: 185 DAC12: 186 DAC2: 235			68.98	8501.71	10590.64	260.71
		S40	1.5°C	SSP5	3.00%			DAC11: 78 DAC12: 101 DAC2: 137			11904.54	30422.17	30657.64	1458.17
		S41	2°C	SSP5	3.00%						4746.87	31248.17	31565.86	1251.30
		S42	1.5°C	SSP1	3.00%			DAC11: 296-185 DAC12: 384-186 DAC2: 402-235	DAC11: 8.1-5.3	DAC11: 1.3 DAC12: 5.5-5 DAC2: 2.5	395.65	3273.68	1815.59	128.28
		S43	2°C	SSP1	3.00%						10.58	1296.14	4612.53	86.59
		S44	2°C	SSP1	3.00%			DAC11: 296 DAC12: 384 DAC2: 402			0.02	2.13	33.63	0.43
		S45	2°C	SSP1	3.00%			DAC11: 78 DAC12: 101 DAC2: 137			171.96	7623.08	8076.18	243.64
		S46	2°C	SSP2	3.00%			DAC11: 185 DAC12: 186 DAC2: 235	DAC11: 5.3	DAC11: 1.3 DAC12: 5 DAC2: 2.5	247.47	9792.29	9475.86	338.24
		S47	1.5°C	SSP4	3.00%			DAC11: 78 DAC12: 101 DAC2: 137			4303.03	21486.02	17655.80	902.74
		S48	2°C	SSP4	3.00%						688.73	20635.53	21033.43	661.84
Grant et al. (2021)	TIAM-Grantham	S49	1.75 °C	SSP2	1.00%		HT-DAC (DAC1) and LT-DAC (DAC2)	DAC1: 300-100 DAC2: 350-50	DAC1: 8.1 DAC2: 7.2	DAC1: 1.8 DAC2: 1.2	15.19	1065.29	10177.08	118.37
		S50	1.75 °C	SSP2	3.00%						15.19	1065.29	20505.29	180.34
		S51	1.75 °C	SSP2	5.00%						15.19	1065.29	20531.20	180.38
		S52	2°C	SSP2	1.00%						15.19	1065.29	10112.71	117.99
		S53	2°C	SSP2	3.00%						15.19	1065.29	16571.45	156.62
		S54	2°C	SSP2	5.00%						0.00	1065.29	20483.43	179.93

Table 5
Factors affecting DACCS deployment in different IAM studies.

Category of Change	Shift in Scenario From	Shift in Scenario To	Change in DACCS Deployment (%)
Change in future socio-economic pathway*	S35	S36	-95.90%
	S46	S48	95.67%
	S41	S46	-72.97%
	S41	S43	-93.08%
	S43	S46	290.63%
	S40	S42	-91.20%
	S40	S47	-38.09%
	S42	S47	603.73%
Increased temperature target	S1	S2	-47.31%
	S5	S8	-73.97%
	S6	S9	-29.37%
	S7	S10	-100.00%
	S11	S13	-13.10%
	S12	S14	-1.22%
	S33	S34	-93.15%
	S31	S32	-94.22%
	S29	S30	-88.16%
	S35	S37	-99.29%
	S42	S43	-32.50%
	S40	S41	-14.19%
	S47	S48	-26.68%
	S49	S52	-0.33%
	S50	S53	-13.15%
	S51	S54	-0.25%
Decreased Overshoot	S24	S26	-42.38%
	S25	S27	-30.71%

Category of Change	Shift in Scenario From	Shift in Scenario To	Change in DACCS Deployment (%)
Decreased Discount Rate	S11	S21	-13.79%
	S51	S50	-0.02%
	S51	S49	-34.38%
	S50	S49	-34.36%
	S54	S53	-12.96%
	S53	S52	-24.67%
	S54	S52	-34.43%
Limited Total CDR	S6	S7	-93.97%
	S6	S8	-96.57%
	S9	S8	-95.14%
	S9	S10	-100.00%
	S5	S7	-54.20%
	S8	S10	-100.00%
Limited DACCS	S11	S16	-85.63%
Decreased DACCS grwoth rate limit	S11	S18	-56.32%
	S19	S11	-15.53%
	S19	S18	-63.11%
	S17	S16	-10.71%
	S16	S15	-20.00%
Reduced Cost of DACCS	S11	S20	1.15%
	S24	S25	47.73%
	S26	S27	77.67%
	S38	S39	7670.99%
	S44	S45	57140.09%
Change in Carbon Price Trajectory	S29	S28	-43.26%
	S29	S31	-17.20%
	S29	S33	-99.44%

* Changes in the shared socio-economic pathways (SSPs) for scenarios by Furhaman et al. (2021) also change the costs and energy requirements of DACCS.

Red and Green colour codes are indicating the the intensity (relative) of negative and positive changes in DACCS cumulative deployment.

870 to 750 GtCO2 [28]. Fuhrman et al. (2021) used the SDR of 3% [29]. Grant et al. (2021) considered three scenarios of 1, 3 and 5% SDR and observed the decreasing effects of lower SDR values in CDR deployments. DACCS deployment (in 2100) was approximately doubled when SDR was set to 5% compared to the initial value of 1% both for 1.75 and 2 °C scenarios. In addition to the effects of SDR changes on CDR deployment, their results also show that using higher SDR values leads to higher 2030 CO_2 emissions, lower yearly mitigation investments, lower renewables deployment, lower energy demand reduction in 2030 and a higher share of fossil fuel in 2030 final energy [63].

In many scenarios, it is contemplated that temperature rise will pass the threshold of 1.5 °C some years before 2100. Potentially, through applications of CDR, the change in global average temperature could be returned to below 1.5 °C at the end of the century — a phenomenon referred to as the "overshoot" of the global temperature goal. Due to the risks associated with triggering tipping points or irreversible climate change impacts caused by passing 1.5 °C [64–71], limiting the magnitude and duration of overshoot as much as possible 72 is advisable. Preventing the model from experiencing overshoot (especially for the 1.5 °C goal) will increase mitigation costs and require higher rates of emissions reductions with less CDR [3]. CDR methods can, at best, compensate for emissions released over the remaining carbon budget, but they cannot alleviate the burdens of irreversible biophysical changes. Fuhrman et al. (2020) investigated restricting overshoot (from high to low) for two high and low-cost scenarios (S24 and S25, respectively). By constraining global CO_2 emissions pathways, they derived two global warming trajectories of (1) high-overshoot and (2) low-overshoot, both meeting the end-of-century temperature target of 1.5 °C. The former will experience a global warming level of 1.78 °C in 2055, followed by a rapid decline till 2100. In the latter, the peak temperature is 1.56 °C, which occurs in 2045, and the end-of-century warning level is 1.32 °C. When changing the condition from high to low overshoot, they found a significant drop in the cumulative DACCS deployment (around 30% to 40% reduction) [18]. These changes are shown in the left columns of Table 5 for the category "Decreased Overshoot" when shifting scenarios from S24 to S26 and S25 to S27. Moreover, increasing the temperature target (e.g., from 1.5 to 2 °C) decreases the cumulative DACCS deployment in many scenarios (see the left side of Table 5). Since the remaining carbon budget corresponding to a lower temperature target is also lower, models use more CDR to compensate for emissions. This result is also reflected in the median values of DACCS deployment in the AR6 database scenarios when moving from the C3 and C4 categories (2 °C) to C1 and C2 (1.5 °C), as summarized in Table 2.

3.3. Maximum capacity and growth rate limitations

Some models set limits on CDR/DACCS deployment to avoid models over-relying on such strategies to compensate for the lack of more proven mitigation strategies. This issue is crucial for BECCS and AR, which might threaten sustainability goals when deployed at large scales (Fig. 3). In the peer-reviewed IAM studies examined for this article, Holz et al. (2018) used 50% of the potential of different CDR methods, assumed by En-ROADS and C-ROADS, as limits. This corresponds to 2.5 $GtCO_2$/year and 3.5 $GtCO_2$/year for BECCS and DACCS, respectively. Strefler et al. (2018) released the results of CDR deployment in six scenarios consisting of different combinations of two temperature targets of 2 and 1.5 °C and three upper limitations on yearly total CDR deployment of 8, 12 and 20 $GtCO_2$. For 1.5 °C scenarios, the DACCS deployment rates in 2100 were 2.1 (S7), 3.5 (S5) and 10.7 (S6) $GtCO_2$/$year_{2100}$ for 8, 12 and 20 GtCO2/year maximum CDR deployment, respectively [27] (shown in Table 4). This means that even limiting the total CDR deployment to 8 $GtCO_2$, at least 25% of the removed carbon comes from DACCS, which could be a proper conservative estimate of possible DACCS deployment in 1.5 °C scenarios. However, when the temperature target is 2 °C, and there is an 8 $GtCO_2$/year limit on the total CDR deployment, DACCS will not be deployed (S10) [27]. Realmonte et al. (2019) assumed 10 Gt CO_2/year for BECCS deployment for all scenarios and investigated the DACCS deployment changes when its role is limited to 30 and 3 $GtCO_2$/year, corresponding to scaling the sorbent production market by a factor of 100 and 10. Their study also incorporated the constraints on the possible growth rate of DACCS to enhance the feasibility of their pathways [28]. Some studies criticize IAMs because of their infeasible speed of technological change [72]. This parameter might be hard to determine as it depends on various factors, such as modularity, political and economic incentives and public acceptance of DACCS, which could also vary for different SSPs. Realmonte et al. (2019) used historical diffusion rates of various technologies, such as wind, solar, natural gas, and nuclear energy, and decided to use three values of 15, 20 (baseline) and 30% as growth rates that might apply to DACCS [28]. The significant impacts of these changes (decreased DACCS growth rate limit) on the DACCS cumulative deployment are shown in Table 5. Other IAM studies reviewed here have not analyzed the temporal deployment of DACCS, which is a crucial factor for producing pathways that might be more plausible and informative for real-world settings.

3.4. Non-Energy costs of DACCS

DACCS deployment in IAMs is limited by its costs. The technology is not mature, and the cost estimations from techno-economic analyses and a limited number of commercial projects vary substantially. Such uncertainty makes reaching a consensus on the costs of DACCS very challenging. This has led some studies to do a sensitivity analysis of the costs associated with DACCS. Fuhrman et al. (2019) investigated four different scenarios meeting 1.5 °C by combining two levels of overshoot (high and low) and costs of DACCS (high and low estimates from the literature). Fixing the overshoot variable shows that a decrease in unit costs from 300 $/t$CO_2$ to 180 $/t$CO_2$ increases the cumulative deployment of DACCS from 863 to 1274 GtCO_2 for high overshoot scenarios (S24 and S25) and from 497 to 883 GtCO_2 for low overshoot scenarios (S26 and S27). Realmonte et al. (2019) also found that when the initial cost of DACCS changed from 300 to 180 $/t$CO_2$ for HT-DAC and 350 to 100 $/tCO2 for LT-DAC, it increased the cumulative deployment rate by around 10 GtCO_2. Since the models and their other assumptions are different, it is hard to determine why the model used by Fuhrman et al. (2019) showed more sensitivity compared to Realmonte et al. (2019). However, based on the available information, the model Realmonte et al. (2019) developed has two prominent elements that might have contributed to their findings. First, they put an upper limit on DACCS deployment of 30 GtCO2 (which was already reached even in the high-cost DACCS scenario). Second, the costs of DACCS in their model were not fixed, and they provided floor costs of DACCS, which were 100 (HT-DAC) and 50 $/t$CO_2$ (LT-DAC) for both high and low-cost scenarios [28]. Floor costs are the minimum cost the technology might achieve with the learning rate in the coming years. All the costs mentioned above exclude the energy costs enabling the model to incorporate it endogenously depending on the energy source cost trajectory [28,29]. For this reason, estimates of total costs are higher than the non-energy costs of DAC. For example, the International Energy Agency (IEA) GHG estimates that the cost of first-of-a-kind Direct Air Capture and Carbon Storage (DACCS) technology ranges from $400 to $700 for a net removal of one MtCO2, incorporating the emissions of the energy source, using global average solar photovoltaics (PV) [73].

Another critical factor is how DACCS costs evolve, which depends on many uncertain socio-economic, technological, and political factors and the DACCS deployment [74]. While some IAM studies use a fixed cost for the simulation period, a group of studies has differentiated between the initial cost and the floor cost [26,28,29,75]. The numerical representation of the learning rate shows the percentage reduction of costs at each time step that the cumulative capacity doubles [76]. While the median learning rate of solar PV (photovoltaic) is 23% (from 1959 to 2011) and for onshore/offshore wind is 12% (from 1979 to 2010), much lower rates are observed in technologies such as geothermal, nuclear, and hydroelectricity [77]. Predicting the learning rate for DACCS is a highly uncertain process. Fasihi et al. (2019) used 10% and 15% values as the conservative and baseline estimates of the learning rate variable for DACCS [78]. These learning rates were derived from the literature on DACCS and similar technologies [79,80]. Finally, carbon pricing may also affect the scale of DACCS deployment due to path dependencies that might be introduced by cheaper CDR technologies (e.g., BECCS, enhanced weathering). This issue was examined in studies by Strefler et al. and is discussed further in the supplementary information (Figure S3).

4. Modeling recommendations to explore and better understand the scale and pace of direct air capture deployment

4.1. Regional differences and the need for Context-based regional scenarios

Across the scenarios reviewed for this article, there are significant regional differences in DACCS deployment, even for scenarios with the same SSP, temperature target and range of DACCS deployment. Fig. 4 compares four scenarios of S1, S6, S26 and S27, respectively, removing 580, 361, 479 and 883 GtCO_2 by DACCS for the whole century. DACCS in the Middle East and North Africa provide 25% and 20% of total removals in the S1 (MERGE-ETL used by Marcucci et al. 2017) and S6 (REMIND model used by Strefler et al. 2018) scenarios, while no DACCS is being deployed in these regions in scenarios S26 and S27 (GCAM results by Fuhrman et al. 2020). Worldwide, current DAC pilot projects are in the developed economies of North American countries (by Carbon Engineering) and European countries such as Iceland and Switzerland (by Climeworks). A significant open question is how countries with fewer financial and technological resources, institutional capacity and political stability might deploy this expensive technology on the scales that some IAMs project. For this reason, *we recommend the integration of global IAMs with national analyses to deliver more realistic estimates of different pathways that might meet global climate change targets.*

Geographical differences might lead to different deployment pathways for DACCS and DACCU. Geological storage options and available capacity vary among regions. In addition to the availability of reservoirs, the economic and technical feasibility of storage, public acceptance [81], and regulatory frameworks might be in favor or against storage. Moreover, regional demand for CO_2 as a raw material and/or an international market for CO2-based products may significantly promote DACCU in different regions [82].

Downscaled or extended SSP scenarios are appropriate tools for analyzing different climate mitigation, adaptation and CDR strategies and policies at national and sub-national levels [83,84]. Such scenarios can be valuable in different ways:

- Being informative for assessing different mitigation policies and leading to more reliable and realistic regional outcomes with varying characteristics of regions that may influence the performance and efficiency of CDR/DAC alternatives [83,85].
- Helping policy-makers analyze their mitigation goals or international pledges (such as nationally determined contributions reported to the United Nations based on the Paris Agreement) and facilitating pathways toward meeting their targets [86,87].

4.2. Modeling both high-temperature (HT) and low-temperature (LT) DAC

In addition to the benefits of modeling different CDR methods and having a diverse CDR portfolio, having both HT-DAC and LT-DAC needs to be a more typical exercise among studies. Such considerations are important because the system demands of these two approaches vary significantly (shown in Table 1), which can, in turn, shape different temporal and spatial deployment pathways for them. While HT-DAC and LT-DAC are mature technologies proven to be techno-economically feasible, HT-DAC seems more mature, as it utilizes the technology and

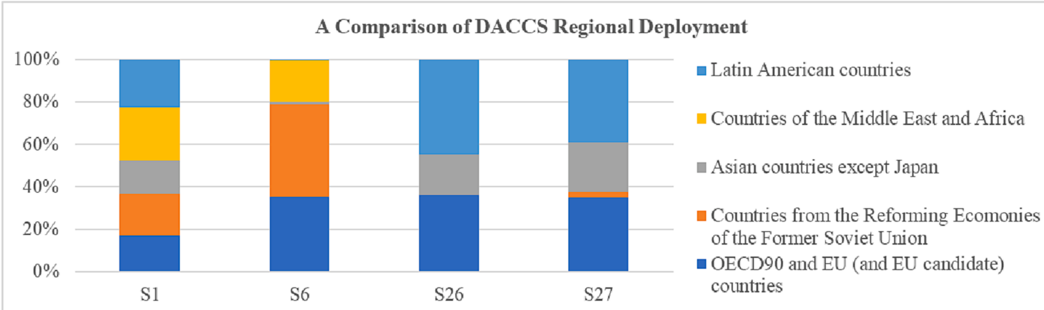

Fig. 4. Variations in the regional deployment of DACCS among scenarios.

equipment used in other industries (namely pulp and paper) [28]. This could be the reason why many reviewed studies, such as Marcucci et al. (2017), Holz et al. (2018), Strefler et al. (2018), Fuhrman et al. (2020), and Strefler et al. (2021a and 2020b) only incorporated HT-DAC. Three studies [28,29,63] reviewed in this paper (from a total of nine) considered both variants of DACCS. According to their results, *we recommend having both options available in models*. One of the critical differences between them is energy requirement. Since LT-DAC needs a relatively lower temperature (around 80 to 120 °C) of heat energy compared to HT-DAC (above 800 °C), the required heat energy can be supplied from different sources, such as industrial low-grade waste heat, renewables, heat pumps and geothermal [22,41,78,88–90]. On the other hand, 800–900 °C thermal energy cannot be provided by using second-hand energy [89], which makes natural gas a desirable energy supply option.

In most of the scenarios in the study by Realmonte et al. (2019), the LT-DAC contribution is higher than HT-DAC, probably due to the costs and energy requirements of HT-DAC (Table 1). They found that HT-DAC will be deployed later than LT-DAC because cheaper natural gas is anticipated to be available later in the century due to less fossil fuel being used for producing electricity (Realmonte et al., 2019). While in most of the scenarios of Fuhrman et al. (2021), HT-DAC and LT-DAC were found to be deployed at the same scale (S46), there were other scenarios where either HT-DAC (S41, S47 and S48) or LT-DAC (S42 and S43) was the major DAC variant being deployed. These differences are driven in part by specific characteristics about alternative socio-economic futures. When renewables and nature-based solutions are favored and there is less dependency on fossil fuels (SSP1), LT-DAC has higher deployment. In such scenarios, renewables are available at lower costs, and since LT-DAC needs low-temperature heat energy, thermal energy for DAC can be produced by electric heat pumps. On the other hand, HT-DAC had a more significant role in scenarios corresponding to worlds with a higher reliance on fossil fuels (SSP4 and SSP5). This is because the high-temperature heat required for HT-DAC is provided by cheap natural gas with carbon capture and storage [29].

4.3. The necessity of considering both carbon utilization and storage pathways

Most IAM studies do not incorporate both CO_2 capture options of utilization and storage after deploying DAC technology. In DACCS, the captured CO_2 will be permanently sequestered in geological reservoirs; in DACCU, it will be utilized for different purposes, such as chemical feedstock (e.g., material and fuel synthesis) and injection fluid [23,24]. The implications of these two options for the carbon balance are very different. While introducing and advancing "clean fuels" is necessary for mitigating climate change by substituting fossil fuels, decarbonizing electricity, and reducing the need for a large amount of biomass with land-use problems [82], a DACCU system should not be considered "negative" from a carbon balance standpoint. Fuels produced using captured carbon can be, at best, carbon neutral due to emissions released during their use phase [23]. Studies that examined DACCU systems have primarily focused on life cycle assessments, techno-economic analysis and possible projections of the future role of DACCU. Galimova et al. (2022) depicted a 100% renewable energy system that requires DACCU at a scale of 4.8 $GtCO2/year_{2050}$ for producing fuels. However, such high-range estimates might be limited when IAMs are used to investigate the DACCU deployment by capturing systematic interactions. A good example can be a study by Akimoto et al. (2021), where they analyzed the complex interactions between DAC and utilization systems (regarding the marginal abatement cost of CO_2) and projected a carbon utilization market of 1 $GtCO2/year_{2100}$ in a scenario meeting a 1.5 °C target [24]. Since IAMs aim to model interactions between different economic, social and energy sectors and can incorporate various shared socio-economic uncertainties, *we recommend that IAM studies explore how the temporal and spatial deployment of DACCU and DACCS might materialize under different climate change targets, SSPs, assumptions regarding carbon utilization markets, and costs of technologies*.

Furthermore, DACCS deployment is only possible if a government policy regarding emissions reduction or carbon removal (e.g., carbon tax, cap-and-trade system) is in place. A carbon utilization market would alleviate the dependency of DACCS on government policy [91]. Moreover, the availability of DACCU in the early stages of DAC deployment decreases the burdens of high contemporary costs of DAC technology, providing an opportunity for early projects to grow faster and allowing for economies of scale to reduce costs via learning-by-doing [74]. We can also see that various companies, such as Climeworks, Carbon Engineering and Global Thermostat, are deploying DACCU or have plans for demonstration projects [92]. This observation also highlights the necessity for models to consider utilization pathways of the captured carbon by DAC.

Finally, because the current level of carbon storage is negligible and might lack public support and regulatory frameworks, *we recommend that IAM studies use conservative estimates of regional available geological storage and/or perform sensitivity analysis on this parameter*.

4.4. Differences between models, underlying assumptions, and the need for model intercomparison projects

IAMs explore the interactions between social and natural systems to assist policy-makers, provide insights for the general public, and contribute to scientific analysis [93,94]. While these models can help assess the impacts of different policies or find pathways to meet specific targets, a lack of transparency in communicating modeled assumptions and uncertainties or not having the proper knowledge to interpret the results can be misleading. *We recommend that modeling studies should be transparent about the underlying assumptions that affect DACCS deployment and that multi-model assessments be produced* (e.g., *model intercomparison projects) to analyze the robustness of results, making them valuable for policy-making.*

The results for DAC deployment of different modeling activities vary significantly, even when they try to model pathways to the same target (e.g., scenarios meeting 1.5 °C) or assess the effects of the same policies (e.g., studies that explore the outcomes of countries' pledges). We showed DACCS deployment covers a wide range between the cumulative deployment of 0.4 $GtCO_2$ to around 1200 $GtCO_2$ (in scenarios that meet the 1.5C° target). The reason behind this variation can be classified into two categories: 1) research objectives (e.g., modeling pathways with low energy demand) and their underlying assumptions [95] and 2) differences in the structures of the IAMs [96,97].

Most of our focus in this paper was on modeling assumptions, but we also discussed some issues regarding the structure of the model (see different CDR methods presented in models in section 2). Being transparent about the modeling assumptions and the research objectives can help address and interpret this source of uncertainty. Due to the various modeling assumptions and structures employed, comparing the results from different modeling exercises to completely characterize the causal relations between the drivers of a specific scenario or model element and the modeled outcomes is impossible. For scenarios reviewed here, Fig S4 (in the supplementary information) summarizes the differences between various projected energy sources and projected final energy demand variations. Among SSP2 pathways that meet 1.5 °C, we see scenarios heavily relying on nuclear, coal, oil and gas energy supply in 2100 (S24 to S27 by Fuhrman et al. [2020]), and others showing a significant expansion of renewables such as wind, hydro, geothermal and solar (S28 to S24 by Strefler et al. [2021a]). As discussed previously, such differences will also affect how much CDR/DACCS is required as well as the distribution of HT- and LT-DAC.

A proposal for enabling the inter-comparison of DAC projections among different IAMs is "harmonization" [98] which should include socio-economic assumptions, techno-economic assumptions, economic aspects related to fossil fuel prices, and climate-related implementation policies [95]. We believe the SSP family scenarios have successfully addressed many of these issues, specifically for socio-economic assumptions, and provided a comprehensive, qualitative picture of other aspects [36]. However, studies mostly use only their quantification schemes due to a lack of more specific guidelines regarding model implementation.

The impacts of IAM structure on the modeling results can only be assessed when the modelers have set homogenous assumptions (to the level that structure allows) among models. This is where inter-model comparison studies can be conducted, which could be valuable for enhancing the robustness of results [97]. IAMs might differ in various aspects, such as the representation of energy technology details (also regional distribution), sectoral supply and demand details, availability of mitigation options, incorporation of land use, the inclusion of macroeconomic feedback and type of solution algorithm [93,96,97]. Some studies have investigated the impacts of IAMs' structural differences on specific outcomes, such as BECCS deployment [99] and food security [100] in mitigation scenarios, and have found significant variations in their results. Regarding DACCS, Realmonte et al. (2019) have investigated such effects by analyzing DACCS deployment with two models: TIAM-Grantham [101] and WITCH [102]. Their results of cumulative DACCS deployment for both models were in the same range for baseline scenarios. However, more variations in DACCS deployment were observed in some of the sensitivity analysis scenarios. Additionally, the electricity energy mix (contribution of different sources) and BECCS and AR deployment are pretty diverse among the models (shown in the supplementary figure S4 of Realmonte et al., 2019), further illustrating the impact of model structure on outcomes.

5. Discussion and conclusion

This article focused on IAM projections of DACCS and discussed the nuances behind significant variations of its deployment pathways in different scenarios to achieving Paris Agreement targets. We investigated the IPCC AR6 database in addition to a more in-depth analysis of 54 IAM scenarios published in the peer-reviewed literature modeling DACCS [18,26–33]. We began by noting that two technological variants of DAC, HT and LT, have significantly different resource demands for energy and water, so it is essential to explore how much DAC might be demanded to achieve Paris targets and how many resources DAC may require. Such questions are appropriate to explore with IAMs. We then investigated the IPCC AR6 database to see how well DAC was represented among scenarios that achieve Paris Agreement targets. DAC is covered in a minority of relevant AR6 scenarios (253 of 700).

Before drawing conclusions from the AR6 database scenarios for DAC, we examined the larger context of CDR demand in the scenarios. Amongst various scenario factors that might matter for differences in CDR deployment, we observed that fossil fuel energy supply was the only factor that distinguished scenarios with low cumulative CDR from high cumulative CDR. We also found that a majority of AR6 database scenarios deploy AR and BECCS at levels that exceed sustainable deployment thresholds (60% of scenarios and 85% of scenarios, respectively). Such findings are concerning since a high reliance on land-intensive CDR methods such as BECCS has been controversial [47,103] because of its possible adverse effects on food availability and prices. From the AR6 database scenarios that include DACCS, we found a reduction in the average (around 5%) and median (around 15%) of BECCS deployment in scenarios with DACCS as an option (see Figure S1 in the supplementary information).

The AR6 scenario database only hosts the outputs of scenario studies, and the underlying assumptions (including study design and model structure) of IAM studies matter. We, therefore, consulted peer-reviewed studies of DACCS deployment to capture such dimensions that cannot be gleaned from the AR6 database alone [18,26–33]. Noteworthy factors were aggregated into four categories: 1) uncertainties regarding socio-economic futures (Section 3.1), 2) intergenerational assumptions (Section 3.2), 3) maximum capacity and growth rate limitations (Section 3.3) and 4) costs of DACCS (Section 3.4). Table 5 summarizes our key findings.

First, with respect to socio-economic assumptions, we found that most scenarios (41 out of 49 reviewed here) use the assumptions of SSP2, even though DACCS deployment will significantly change under different SSPs. For example, with a shift from SSP1 to SSP4 (e.g. a shift from S42 to S47), DACCS deployment increases by around 600%. This example can be found in Table 5 for the category "change in the future socio-economic pathway". The elements deriving these behavioral changes in models include but are not limited to demographic characteristics (e.g., GDP and population projections as summarized in Fig. 1), technological change and innovation trends, and the level of respect toward environmental boundaries [36]. As there is no guarantee of what path the world will choose for the rest of the century, conducting studies assessing DACCS (or other mitigation policies) under various levels of socio-economic challenges embodied in the narratives of different SSPs would be helpful.

Second, regarding intergenerational assumptions, temperature targets and overshoot levels affect DACCS deployment as summarized in Table 5. For the categories "increased temperature target" or "decreased overshoot" level the need for DACCS may be significantly reduced. In some scenarios, a change in temperature target from 1.5 to 2 °C reduces DACCS deployment by −100% (e.g., a shift from S7 to S10), which means DACCS is not being deployed anymore. Another important factor is the social discount rate (SDR), which is an economic parameter that models use to estimate the present value of future costs and benefits. This variable is essential in determining the stringency of near-term climate change actions and the level of DACCS/CDR needed in the late century. In the category "decreased discount rate", we see some shifts in scenarios that lead to reductions of up to 35% in DACCS deployment when an SDR of 5% (applied in scenarios S54 and S51) changes to 1% (applied in scenarios S52 and S49).

Third, when models limit the yearly deployment of total CDR or specific methods, such as BECCS and DACCS, their deployment pathways will change. For the category "limited total CDR", there is a 100%

reduction in DACCS deployment when the limit on CDR decreases from 20 to 8 GtCO2/year (e.g., a shift from S8 to S10). Another factor missing in most studies is the growth rate limit that constrains the annual pace at which DACCS deployment can be increased. Table 5 shows "decreased DACCS growth rate limit" can reduce DACCS deployment by 56% when the growth rate limit decreases from 20 to 15 % per year (S11 to S18).

Fourth, there is an extensive range of cost estimates for DACCS used across models, which shows a lack of consensus on both the current and future costs of this technology. By looking at the category of "reduced costs of DACCS" in Table 5, we can observe in most cases a significant increase in DACCS deployment as a result of decreased costs (costs are shown in Table 4). In some cases, DACCS cumulative deployment increases by about 500 times when costs fall by a factor of three (a shift from S44 to S45). Considering the uncertainty around the current cost estimates of DACCS and the future demand for the technology, models are encouraged to do sensitivity analysis around costs to capture this source of uncertainty.

After examining how various modeling assumptions lead to significant variations in projections of DACCS, we also made four recommendations for how IAMs might produce pathways that would be more informative for real-world settings. First, we noted that the projected regional deployment rates of DACCS among models (Fig. 4) are not uniform, so there is also variation in the possible spatial contribution of DACCS globally. To provide policy-relevant insights to national governments, we recommend integrating national/regional perspectives with global IAMs to enhance the feasibility of scenarios.

Second, we reiterated that the system demands of HT-DAC and LT-DAC vary significantly (shown in Table 1), which justifies the need for considering both variants in modeling studies. However, we found only two IAM studies [28,29] that investigated the implications of having both variants of DACCS available as options. The studies suggest that LT-DAC would have higher deployment in worlds with socio-economic outcomes resembling SSP1, where there is less dependency on fossil fuels (including lower cumulative emissions) and renewables make up a large share of the global energy supply.

Third, most IAM studies do not model CO_2 utilization after deploying DAC technology (i.e., DACCU). We discuss notable findings in the literature that suggest CO_2 utilization markets on the order of 1GtCO2/year could be realized as early as 2050. Developments in DACCU are essential to the model because they might alleviate the dependency of DACCS on government policy and provide market incentives for reducing DAC costs via learning by doing.

Fourth, this review cannot provide definitive reasons for the changes in DACCS deployment among different IAM studies since not only do the scenario assumptions differ among studies but also the structures of the models are distinct (discussed in section 4.4). We recommend that future studies investigate the roles of each source of uncertainty in DACCS deployment by analyzing harmonized scenarios under different models (so-called inter/multi-model comparison studies) to produce more robust results.

In addition to the uncertainties observable from the IAMs modeling DACCS, many scholars have criticized some aspects of IAMs that might also contribute to our discussion. These critics revolve around representing heterogeneous actors and aligning with behavioral realism [104–106], technology innovation, diffusion and dynamics [96,105], capital markets and finance [105], carbon pricing and its relationship with socio-technical transitions [105], energy-economy feedback [105], and consideration of ethics and justice [107,108].

While IAMs mostly rely on rational decision-making actors (maximizing the monetary utility), real-world decision-making is often more complex and may not conform to the rationality principles [106]. Billions of dollars in investments in DACCS and the significant role of voluntary markets that companies (early adopters) have shaped around the DACCS ecosystem [109] are examples of decisions that do not follow the traditional definition of rationality (agents select the least-price alternative).

While scientific evidence emphasizes the importance of decarbonization and transitioning to a low-carbon lifestyle, carbon dioxide removal (CDR) methods models may overestimate their potential impact due to discounting future mitigation costs and neglecting hard-to-quantify behavioral changes (e.g., diet changes). This has led some researchers to recommend collaboration between ethicists and energy-economy modelers to reduce inherent risks and consider justice principles [107]. There is an equity-driven element to CDR and DACCS provision that also does not get reflected in IAMS. The Paris Agreement is informed by differentiated responsibilities framed as expectations that developed states will contribute to mitigation measures with greater ambition. However, the geographic distribution of CDR opportunities based on biophysical capacities or least-cost models show different outcomes than scenarios that account for equitable considerations, such as the ability to pay or historical responsibility [110,111]. These concerns will be relevant to the trajectory of DACCS development.

CRediT authorship contribution statement

Kasra Motlaghzadeh: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Vanessa Schweizer:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Neil Craik:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Juan Moreno-Cruz:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This is a review paper, and all data that have been used are available in the previous studies and the database mentioned in the paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2023.121485.

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